Mounting Drives in Linux

Contents

Mount a USB in WSL

If you have a windows USB that is E drive and you also want it mounted in WSL Linux

MOUNT USB

sudo mount -t drvfs E: /mnt/e

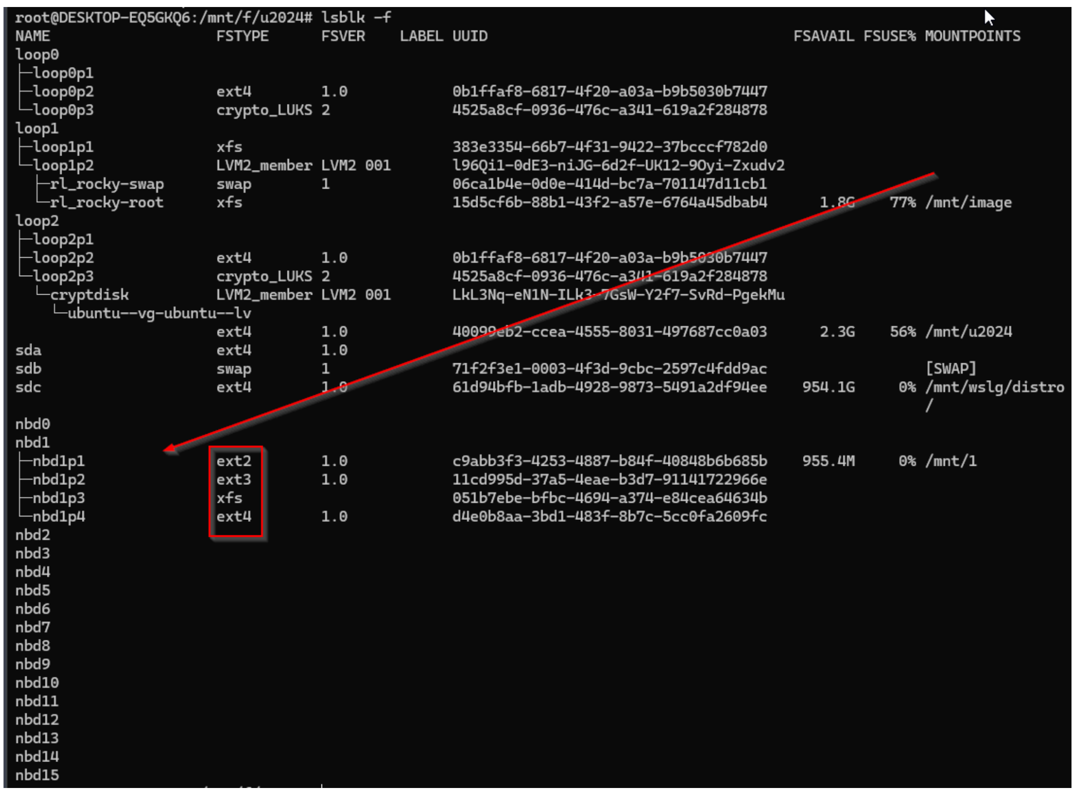
Mount VMDK

sudo apt install qemu-utils

sudo modprobe nbd

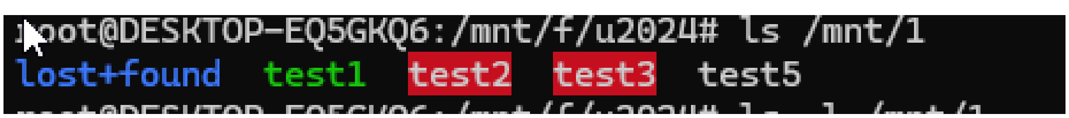
sudo qemu-nbd -r -c /dev/nbd1 U2024_1-flat.vmdk

lsblk -f



```
root@DESKTOP-EQ5GKQ6:/mnt/f/u2024# lsblk -f
NAME                  FSTYPE      FSVER    LABEL UUID                                   FSAVAIL FSUSE% MOUNTPOINTS
loop0
├─loop0p1
├─loop0p2             ext4        1.0            0b1ffaf8-6817-4f20-a03a-b9b5030b7447
└─loop0p3             crypto_LUKS 2              4525a8cf-0936-476c-a341-619a2f284878
loop1
├─loop1p1             xfs                        383e3354-66b7-4f31-9422-37bcccf782d0
└─loop1p2             LVM2_member LVM2 001       l96Qi1-0dE3-niJG-6d2f-UK12-9Oyi-Zxudv2
  ├─rl_rocky-swap     swap        1              06ca1b4e-0d0e-414d-bc7a-701147d11cb1
  └─rl_rocky-root     xfs                        15d5cf6b-88b1-43f2-a57e-6764a45dbab4      1.8G    77% /mnt/image
loop2
├─loop2p1
├─loop2p2             ext4        1.0            0b1ffaf8-6817-4f20-a03a-b9b5030b7447
└─loop2p3             crypto_LUKS 2              4525a8cf-0936-476c-a341-619a2f284878
  └─cryptdisk         LVM2_member LVM2 001       LkL3Nq-eN1N-ILk3-7GsW-Y2f7-SvRd-PgekMu
    └─ubuntu--vg-ubuntu--lv
                      ext4        1.0            40099eb2-ccea-4555-8031-497687cc0a03      2.3G    56% /mnt/u2024
sda                   ext4        1.0
sdb                   swap        1              71f2f3e1-0003-4f3d-9cbc-2597c4fdd9ac                  [SWAP]
sdc                   ext4        1.0            61d94bfb-1adb-4928-9873-5491a2df94ee    954.1G     0% /mnt/wslg/distro
                                                                                                       /
nbd0
nbd1
├─nbd1p1              ext2        1.0            c9abb3f3-4253-4887-b84f-40848b6b685b    955.4M     0% /mnt/1
├─nbd1p2              ext3        1.0            11cd995d-37a5-4eae-b3d7-91141722966e
├─nbd1p3              xfs                        051b7ebe-bfbc-4694-a374-e84cea64634b
└─nbd1p4              ext4        1.0            d4e0b8aa-3bd1-483f-8b7c-5cc0fa2609fc
nbd2
nbd3
nbd4
nbd5
nbd6
nbd7
nbd8
nbd9
nbd10
nbd11
nbd12
nbd13
nbd14
nbd15
```

mount /dev/nbd1p1 -o ro /mnt/1



```
root@DESKTOP-EQ5GKQ6:/mnt/f/u2024# ls /mnt/1
lost+found  test1  test2  test3  test5
```

LVM

apt install lvm2

Logical Volume Management maps physical storage as virtual disks providing more flexibility.

#fdisk -l IMAGENAME

```
root@BWW-HP:/mnt/d/rocky# fdisk -l ROCKY-flat.vmdk
Disk ROCKY-flat.vmdk: 10 GiB, 10737418240 bytes, 20971520 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6a3e87b2

Device            Boot   Start      End  Sectors Size Id Type
ROCKY-flat.vmdk1 *        2048  2099199  2097152   1G 83 Linux
ROCKY-flat.vmdk2       2099200 20971519 18872320   9G 8e Linux LVM
```

sudo losetup -f -P VMImage-flat.vmdk

sudo lsblk -f

```
root@BWW-HP:/mnt/d/rocky# sudo lsblk -f
NAME       FSTYPE      FSVER    LABEL UUID                                   FSAVAIL FSUSE% MOUNTPOINTS
loop0                                                                          18.7G     4% /mnt/windows_mount
loop1
├─loop1p1 xfs                        383e3354-66b7-4f31-9422-37bcccf782d0
└─loop1p2 LVM2_membe LVM2 001        l96Qi1-0dE3-niJG-6d2f-UK12-9Oyi-Zxudv2
  ├─rl_rocky-swap
  │      swap        1               06ca1b4e-0d0e-414d-bc7a-701147d11cb1
  └─rl_rocky-root
         xfs                         15d5cf6b-88b1-43f2-a57e-6764a45dbab4       1.8G    78% /mnt/image
sda        ext4        1.0
sdb        swap        1             ffaa50db-fef5-4d53-874d-703b235959ba                   [SWAP]
sdc        ext4        1.0           26665ad7-a2f3-497e-a01b-84cef2b054bb       946G     1% /mnt/wslg/distro
                                                                                            /
```

Add the Physical device to LVM

root@BWW-HP:/mnt/d/rocky# sudo pvscan --cache
 pvscan[1047] PV /dev/loop1p2 online.
Add volume groups and logical volumes associated with image

root@BWW-HP:/mnt/d/rocky# **sudo vgs**
 VG   #PV #LV #SN Attr   VSize VFree
 rl_rocky   1   2   0 wz--n- <9.00g   0

root@BWW-HP:/mnt/d/rocky# **sudo lvs**
 LV   VG   Attr   LSize Pool Origin Data% Meta% Move Log Cpy%Sync Convert
 root rl_rocky -wi-ao---- <8.00g

swap rl_rocky -wi-a----- 1.00g

Activate the volume group and LVs:

root@BWW-HP:/mnt/d/rocky# **sudo vgchange -ay**
2 logical volume(s) in volume group "rl_rocky" now active

root@BWW-HP:/mnt/d/rocky**# ls -l /dev/mapper**
total 0
crw------- 1 root root 10, 236 Nov 17 11:44 control
lrwxrwxrwx 1 root root 7 Nov 17 19:14 rl_rocky-root -> ../dm-1
lrwxrwxrwx 1 root root 7 Nov 17 19:14 rl_rocky-swap -> ../dm-0

Make a mount point

root@BWW-HP:/mnt/d/rocky# **mkdir /mnt/image**
mkdir: cannot create directory '/mnt/image': File exists

Mount the LV drive

root@BWW-HP:/mnt/d/rocky**# mount /dev/mapper/rl_rocky-root -o ro /mnt/image**

ls /mnt/image

```
root@BWW-HP:/mnt/d/rocky# ls /mnt/image
afs  bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
```

Remove
root@BWW-HP:/mnt/d/rocky# **umount /dev/mapper/rl_rocky-root**
root@BWW-HP:/mnt/d/rocky# **lvchange -an /dev/mapper/rl_rocky-root**
root@BWW-HP:/mnt/d/rocky# **losetup -d /dev/loop1**
root@BWW-HP:/mnt/d/rocky# **pvscan --cache**

LUKS

LUKS is the standard for Linux disk encryption. By providing a standardized on-disk format.

sudo apt-get install cryptsetup

If you see a LUKS partition hope you can find the passphrase.

Identifying a LUKS partition

root@BWW-HP:/mnt/d/u2024# **fdisk -l U2024-flat.vmdk**

```
root@BWW-HP:/mnt/d/u2024# fdisk -l U2024-flat.vmdk
Disk U2024-flat.vmdk: 8 GiB, 8589934592 bytes, 16777216 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 3280AFF7-7E4E-459A-8719-26CD4A789105

Device               Start      End  Sectors  Size Type
U2024-flat.vmdk1      2048     4095     2048    1M BIOS boot
U2024-flat.vmdk2      4096  3674111  3670016  1.8G Linux filesystem
U2024-flat.vmdk3 3674112 16775167 13101056  6.2G Linux filesystem
root@BWW-HP:/mnt/d/u2024#
```

root@BWW-HP:/mnt/d/u2024# **fsstat -o 4096 U2024-flat.vmdk |head -15**

```
root@BWW-HP:/mnt/d/u2024# fsstat -o  4096 U2024-flat.vmdk |head -15
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: Ext4
Volume Name:
Volume ID: 47740b03b5b93aa0204f1768f8fa1f0b

Last Written at: 2024-11-13 13:45:39 (EST)
Last Checked at: 2024-11-13 10:31:09 (EST)

Last Mounted at: 2024-11-13 13:45:39 (EST)
Unmounted properly
Last mounted on: /boot

Source OS: Linux
Dynamic Structure
root@BWW-HP:/mnt/d/u2024#
```

root@BWW-HP:/mnt/d/u2024# fsstat -o 3674112 U2024-flat.vmdk |head -15

```
root@BWW-HP:/mnt/d/u2024# fsstat -o  3674112 U2024-flat.vmdk |head -15
Encryption detected (LUKS)
```

blkid -t TYPE=crypto_LUKS -o device

sudo losetup -f -P U2024-flat.vmdk

sudo lsblk -f

```
root@BWW-HP:/mnt/d/u2024# sudo lsblk -f
NAME       FSTYPE      FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
loop0
├─loop0p1
├─loop0p2 ext4        1.0         0b1ffaf8-6817-4f20-a03a-b9b5030b7447
└─loop0p3 crypto_LUKS 2           4525a8cf-0936-476c-a341-619a2f284878
loop2
├─loop2p1
├─loop2p2 ext4        1.0         0b1ffaf8-6817-4f20-a03a-b9b5030b7447
└─loop2p3 crypto_LUKS 2           4525a8cf-0936-476c-a341-619a2f284878
sda        ext4        1.0
sdb        swap        1           ffaa50db-fef5-4d53-874d-703b235959ba                [SWAP]
sdc        ext4        1.0         26665ad7-a2f3-497e-a01b-84cef2b054bb    946G     1% /mnt/wslg/distro
                                                                                       /
```

root@BWW-HP:/mnt/d/u2024# **sudo cryptsetup luksOpen /dev/loop2p3 cryptdisk**

```
root@BWW-HP:/mnt/d/u2024# sudo cryptsetup luksOpen /dev/loop2p3 cryptdisk
Enter passphrase for /dev/loop2p3: Enter PASSPHRASE
```

ls -l /dev/mapper

```
root@BWW-HP:/mnt/d/u2024# ls -l /dev/mapper
total 0
crw------- 1 root root 10, 236 Nov 18 14:57 control
lrwxrwxrwx 1 root root       7 Nov 18 15:02 cryptdisk -> ../dm-0
lrwxrwxrwx 1 root root       7 Nov 18 15:02 ubuntu--vg-ubuntu--lv -> ../dm-1
```

sudo mount /dev/mapper/ubuntu--vg-ubuntu--lv /mnt/image

ls -l /mnt/image

```
root@BWW-HP:/mnt/d/u2024# sudo mount /dev/mapper/ubuntu--vg-ubuntu--lv /mnt/image
root@BWW-HP:/mnt/d/u2024# ls -l /mnt/image
total 1038436
lrwxrwxrwx   1 root root          7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x   2 root root       4096 Feb 26  2024 bin.usr-is-merged
drwxr-xr-x   2 root root       4096 Nov 13 10:31 boot
dr-xr-xr-x   2 root root       4096 Aug 27 11:39 cdrom
drwxr-xr-x   4 root root       4096 Aug 27 10:21 dev
drwxr-xr-x 108 root root       4096 Nov 14 12:24 etc
drwxr-xr-x   4 root root       4096 Nov 13 11:35 home
```

Remove Mount

cryptsetup luksClose **cryptdisk**

lvchange -an /dev/mapper/ubuntu--vg-ubuntu--lv

cryptsetup luksClose cryptdisk

umount /mnt/image

Mounting Drives in Linux

Bitlocker example of a drive that has a raw multi-part image

affuse usb-19nov.001 /mnt/aff

fdisk -l /mnt/aff/usb-19nov.001.raw

```
root@BWW-HP:/mnt/d/usb-ntfs# fdisk -l /mnt/aff/usb-19nov.001.raw
Disk /mnt/aff/usb-19nov.001.raw: 57.77 GiB, 62026416128 bytes, 121145344 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x5707f3a1

Device                        Boot    Start       End  Sectors  Size Id Type
/mnt/aff/usb-19nov.001.raw1            2048  40962047 40960000 19.5G  7 HPFS/NTFS/exFAT
/mnt/aff/usb-19nov.001.raw2        40962048 121143295 80181248 38.2G  7 HPFS/NTFS/exFAT
root@BWW-HP:/mnt/d/usb-ntfs#
```

#echo 40962048*512 |bc
20972568576

bdeinfo -o 20972568576 /mnt/aff/usb-19nov.001.raw

bdemount -o 20972568576 -p catsanddogs /mnt/aff/usb-19nov.001.raw /mnt/image raw

mount -t ntfs-3g -o ro /mnt/image/bde1 /mnt/windows_mount/

DO NOT TYPE PASSWORD IF SOMEONE IS WATCHING

Mount vmdk in linux

You need aff tools

apt-get install afflib-tools

#affuse IONJADCONNECT-CLONE-000002-flat.vmdk /mnt/vmdk (this command mounts a raw image)

fdisk -l /mnt/vmdk/IONJADCONNECT-CLONE-000002-flat.vmdk.raw

Disk /mnt/vmdk/IONJADCONNECT-CLONE-000002-flat.vmdk.raw: 100 GiB, 107374182400 bytes, 209715200 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x6a9137af

Device Boot Start End Sectors Size Id Type
/mnt/vmdk/IONJADCONNECT-CLONE-000002-flat.vmdk.raw1 * 2048 718847 716800 350M 7 HPFS/NTFS/exFAT
/mnt/vmdk/IONJADCONNECT-CLONE-000002-flat.vmdk.raw2 718848 209711103 208992256 99.7G 7 HPFS/NTFS/exFATmm
Find the offset

#echo 718848*512 |bc (multiply starting sector times sector size)

368050176

#mount -o ro, norecovery, **loop,offset=368050176 /mnt/vmdk/IONJADCONNECT-CLONE-000002-flat.vmdk.raw /mnt/vmdisk**

Image mounted - Read Only

Mount E01 whole disk NTFS

apt install ewf-tools

#mkdir /mnt/ewf_mount

#mkdir /mnt/windows_mount

This mounts the raw image

#ewfmount DT-CBMZYV1.E01 /mnt/ewf_mount

 # fdisk -l /mnt/ewf_mount/ewf1

```
Disk /mnt/ewf_mount/ewf1: 232.9 GiB, 250059350016 bytes, 488397168 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x76aab9e3

Device               Boot     Start       End   Sectors   Size Id Type
/mnt/ewf_mount/ewf1p1 *        2048   1023999   1021952   499M  7 HPFS/NTFS/exF
/mnt/ewf_mount/ewf1p2       1024000 483520511 482496512 230.1G  7 HPFS/NTFS/exF
/mnt/ewf_mount/ewf1p3     483520512 488394751   4874240   2.3G 27 Hidden NTFS W
```

Do the math to find the offset to mount the image

 # echo 10240000*512 |bc

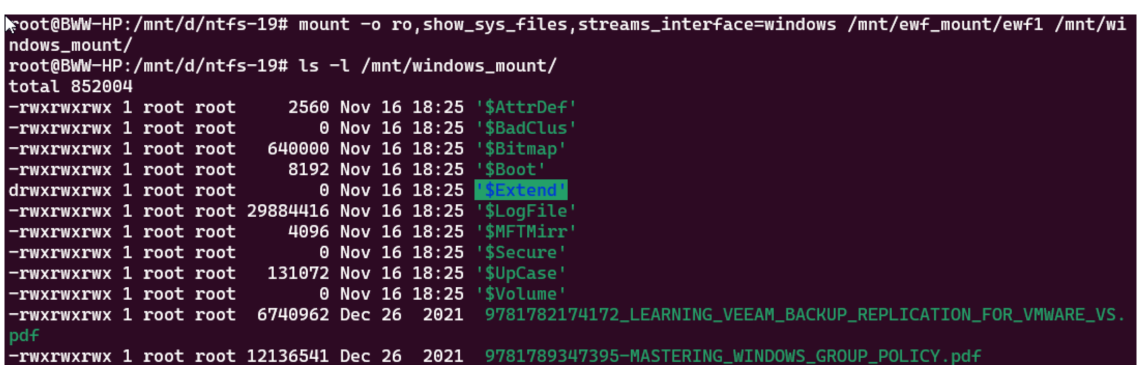
5242880000

#mount -o ro,loop,show_sys_files,streams_interface=windows,offset=524288000 ewf1 /mnt/windows_mount

Mount E01 logical NTFS image

ewfmount /mnt/d/ntfs-19/19-usb.E01 /mnt/ewf_mount/

mount -o ro,show_sys_files,streams_interface=windows /mnt/ewf_mount/ewf1 /mnt/windows_mount/



```
root@BWW-HP:/mnt/d/ntfs-19# mount -o ro,show_sys_files,streams_interface=windows /mnt/ewf_mount/ewf1 /mnt/wi
ndows_mount/
root@BWW-HP:/mnt/d/ntfs-19# ls -l /mnt/windows_mount/
total 852004
-rwxrwxrwx 1 root root     2560 Nov 16 18:25 '$AttrDef'
-rwxrwxrwx 1 root root        0 Nov 16 18:25 '$BadClus'
-rwxrwxrwx 1 root root   640000 Nov 16 18:25 '$Bitmap'
-rwxrwxrwx 1 root root     8192 Nov 16 18:25 '$Boot'
drwxrwxrwx 1 root root        0 Nov 16 18:25 '$Extend'
-rwxrwxrwx 1 root root 29884416 Nov 16 18:25 '$LogFile'
-rwxrwxrwx 1 root root     4096 Nov 16 18:25 '$MFTMirr'
-rwxrwxrwx 1 root root        0 Nov 16 18:25 '$Secure'
-rwxrwxrwx 1 root root   131072 Nov 16 18:25 '$UpCase'
-rwxrwxrwx 1 root root        0 Nov 16 18:25 '$Volume'
-rwxrwxrwx 1 root root  6740962 Dec 26  2021  9781782174172_LEARNING_VEEAM_BACKUP_REPLICATION_FOR_VMWARE_VS.
pdf
-rwxrwxrwx 1 root root 12136541 Dec 26  2021  9781789347395-MASTERING_WINDOWS_GROUP_POLICY.pdf
```